

JEE MAIN REVISION MASTER: METALLURGY

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 10 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Ore Concentration & Pyrometallurgy Extraction	6 PYQs	Metallurgical Processes
Moderate	Ellingham Diagram, Hall-Heroult Process & Refining	4 PYQs	Thermodynamics & Refining

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Ore Concentration & Pyrometallurgy Extraction	$\text{Self Reduction: } \text{Cu}_2\text{S} + 2\text{Cu}_2\text{O} \rightarrow 6\text{Cu} + \text{SO}_2 \quad (\text{Blister Copper})$
Ellingham Diagram, Hall-Heroult Process & Refining	$\text{Hall-Heroult: } 2\text{Al}_2\text{O}_3 + 3\text{C} \rightarrow 4\text{Al} + 3\text{CO}_2$ $\text{Mond: } \text{Ni} + 4\text{CO} \xrightarrow{330\text{K}} \text{Ni(CO)}_4 \xrightarrow{450\text{K}} \text{Ni} + 4\text{CO}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Ore Concentration & Pyrometallurgy Extraction (Moderate Yield — 6 PYQs)

Core Principle: Concentration of sulphide ores by froth flotation: collectors (pine oil, fatty acids, xanthates) make mineral particles water-repellent; depressants (NaCN) separate ZnS from PbS. Roasting converts sulphide ores to oxides with SO₂ gas release. Extraction of copper from copper pyrites involves reverberatory furnace, copper matte (Cu₂S + FeS), and self-reduction in Bessemer converter.

■■ **Caution / JEE Pitfall:** Calcination is heating in absence of air (for carbonates and hydrated oxides); roasting is heating in excess air (for sulphides).

■ **Speed Trick:** Blister copper gets its blistered appearance from the evolution of escaping SO₂ gas during cooling.

Representative Source PYQs:

- [2024 Shift] Q#2 (MCQ): "Select the correct statement :..." [Ans: N/A]
- [2020 Shift] Q#1 (MCQ): "Function of potassium ethyl xanthate in froth floatation process is to make the ore..." [Ans: N/A]
- [2020 Shift] Q#2 (MCQ): "Sulphide ore on roasting gives a gas X. X reacts with Cl₂ in the presence of activated charcoal to give Y. Y is..." [Ans: N/A]

3.2 Ellingham Diagram, Hall-Heroult Process & Refining (Moderate Yield — 4 PYQs)

Core Principle: Ellingham diagram plots ΔG° vs T. A metal can reduce the oxide of any other metal located higher in the diagram. Hall-Heroult process: electrolysis of molten Al_2O_3 with cryolite (Na_3AlF_6) and CaF_2 to lower melting point and increase conductivity. Refining: Zone refining (semiconductors Ge, Si), Mond process for Ni, Van Arkel for Zr and Ti.

■ **Caution / JEE Pitfall:** In Hall-Heroult process, the carbon anodes are consumed by reacting with liberated oxygen to form CO and CO_2 .

■ **Speed Trick:** Zone refining is based on the principle that impurities are more soluble in the molten state than in the solid state of the metal.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "The incorrect statement about Hall -Heroult process is..." [Ans: N/A]
- [2023 Shift] Q#3 (MCQ): "For the formation of which compound in Ellingham diagram ΔG° becomes more and more negative with increase in temperature?..." [Ans: A]
- [2018 Shift] Q#1 (MCQ): "The common impurity present in bauxite is..." [Ans: C]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#1)

The incorrect statement about Hall -Heroult process is

- (A) carbon anode is oxidised to CO and CO_2 .
- (B) Na_3AlF_6 helps to decrease the melting point of the electrolyte.
- (C) CaF_2 helps to increase the conductivity of the electrolyte.
- (D) oxidation state of oxygen changes in the overall cell reaction.

Correct Answer: None

Solution Analysis: The incorrect statement about Hall-Heroult process is given in option (d). It is because in this process oxidation state of oxygen remains same in the overall cell reaction. $2\text{Al}_2\text{O}_3 + 3\text{C} \rightarrow 4\text{Al} + 3\text{CO}_2$

Benchmark Problem 2: KCET 2024 (Q#2)

Select the correct statement :

- (A) Roasting involves heating the ore in the absence of air.
- (B) Calcination involves heating the ore above its melting point.
- (C) Smelting involves heating the ore with suitable reducing agent and flux below its melting point.
- (D) Calcination of calcium carbonate is endothermic.

Correct Answer: None

Solution Analysis: Among the given statements, correct statement is given in option (d). While statement given in options (a), (b) and (c) are incorrect. Their correct form are as follows ■ Roasting involves heating the ore in the regular supply of air. ■ Calcination involves heating the ore below its melting point. ■ Smelting is the process by which a metal is obtained at temperature beyond the melting point from its ore.